Coen Hao

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Physics undergraduate with hands-on experience testing, diagnosing, and validating mechanical and electrical systems.

EDUCATION

University of Toronto *Expected 2028*

Honours B.Sc. — Physics Major; Mathematics Minor; Statistics Minor

Relevant coursework: Practical Physics (Python), Electricity & Magnetism, Classical Mechanics, ODEs, Probability, Statistics & Data Analysis, Linear Algebra

EXPERIENCE

Suspension & Vehicle Dynamics — UTFR Formula Student *2024, 2026 – Present*

- Correlated on-track logged data against MATLAB, Simulink, and Adams Car models to evaluate setups.
- Ran iterative on-track testing and documented results in driver-feedback reports.
- Fabricated chassis jigs and control arms.

EFI Tuning & Powertrain — Independent / Competition Vehicles *2021 – Present*

- Rebuilt 3 full engine and vehicle harnesses from OEM documentation and oscilloscope testing.
- Plotted Haltech data logs in Python to analyze boost, AFR, and intake temperature across competition sessions. Identified a lean AFR under certain ambient conditions and adjusted correction tables accordingly.
- Modeled and optimized intake and exhaust airflow using CFD and 3D-printed prototyping.
- Conducted dyno tuning, drivetrain diagnostics, and electrical diagnosis and repair.

Driver & Vehicle Build Lead — Team Opposite Lock *2021 – Present*

- Designed and integrated a turbocharging system (mechanical, fluid, electrical, and control subsystems) with data logging. Validated reliability over a 1,500-mile road trip and 8+ hours of competition.
- Isolated intermittent idle faults by systematically testing mechanical, electrical, and software causes.
- Built and modified car/tire physics models and Assetto Corsa 3D assets from real-world scans.
- Competed in Final Bout Special Stage Central 2025. Managed team budget, sponsors, and trackside operations.

PROJECTS

ESP32 Fixed-Wing Aircraft — Personal *2026*

- Wrote flight control firmware with a fault-state machine (idle / armed / running / fault) and failsafes for loss of signal. Parsed serial receiver protocol and drove 6 servo channels.

TECHNICAL SKILLS

Data & Programming: Python (SciPy, Matplotlib), MATLAB, Simulink, C/C++ (ESP32 and Arduino), statistical analysis

Test & Measurement: Data logging, oscilloscope and multimeter diagnosis, dyno testing, test documentation & reporting

Engineering & Fabrication: Fusion 360, CFD, suspension kinematics, jig fabrication, wiring harness fabrication